**Table 2. Machine translation on newstest2014.** BLEU and estimated training cost. The Transformer outperforms earlier single models and ensembles at a fraction of their training cost.

| | BLEU ↑ | | Training cost (FLOPs) ↓ | |
|---|---|---|---|---|
| **Model** | **EN–DE** | **EN–FR** | **EN–DE** | **EN–FR** |
| *Single models* | | | | |
| ByteNet [18] | 23.75 | – | – | – |
| Deep-Att + PosUnk [39] | – | 39.2 | – | $1.0 \cdot 10^{20}$ |
| GNMT + RL [38] | 24.6 | 39.92 | $2.3 \cdot 10^{19}$ | $1.4 \cdot 10^{20}$ |
| ConvS2S [9] | 25.16 | 40.46 | $9.6 \cdot 10^{18}$ | $1.5 \cdot 10^{20}$ |
| MoE [32] | 26.03 | 40.56 | $2.0 \cdot 10^{19}$ | $1.2 \cdot 10^{20}$ |
| *Ensembles* | | | | |
| Deep-Att + PosUnk Ensemble [39] | – | 40.4 | – | $8.0 \cdot 10^{20}$ |
| GNMT + RL Ensemble [38] | 26.30 | 41.16 | $1.8 \cdot 10^{20}$ | $1.1 \cdot 10^{21}$ |
| ConvS2S Ensemble [9] | 26.36 | <u>41.29</u> | $7.7 \cdot 10^{19}$ | $1.2 \cdot 10^{21}$ |
| *This paper* | | | | |
| Transformer (base model) | <u>27.3</u> | 38.1 | $3.3 \cdot 10^{18}$ | |
| Transformer (big) | **28.4** | **41.8** | $2.3 \cdot 10^{19}$ | |

Values as reported in the paper. Training cost is the paper's estimate: training time × number of GPUs × sustained single-precision throughput per GPU. Bold: best and underline: second best BLEU in each column, across all rows. Shaded: models from this paper; one training-cost estimate covers both language pairs. – = not reported.